DBMS Tutorial

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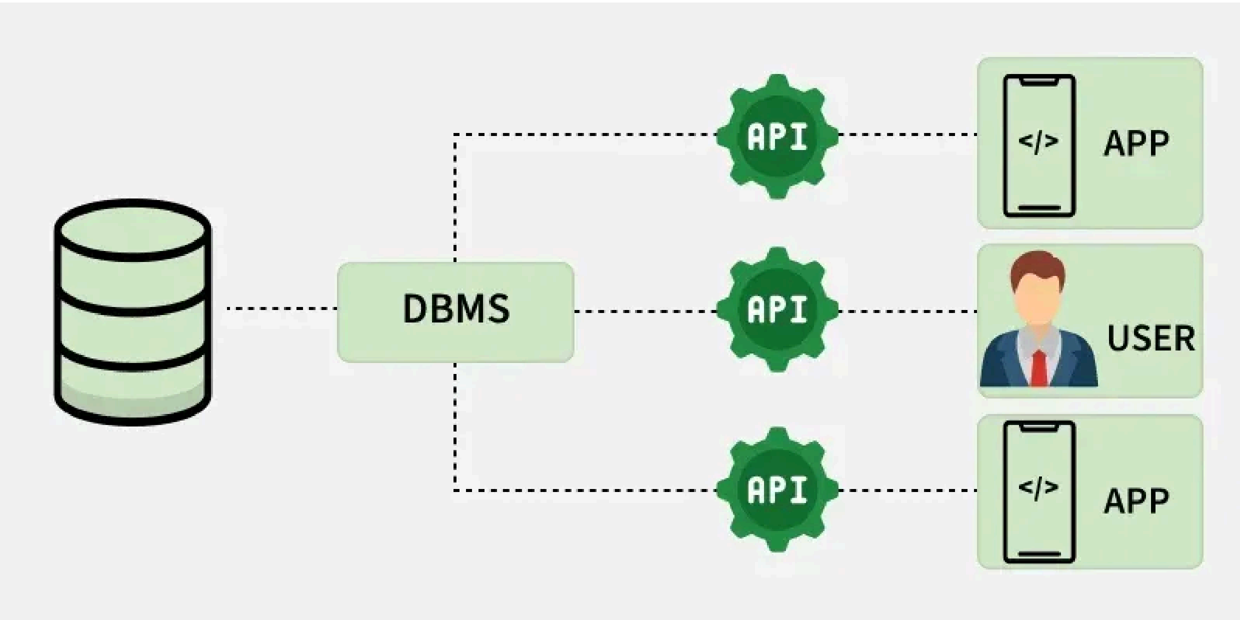
A Database Management System (DBMS) is software used to manage data from a database. It acts as an interface between the database and end users or applications, ensuring data is consistently organised, easily accessible, and secure.

- A database is a structured collection of data that is stored in an electronic device. The data can be text, video, image, or any other format.
- A relational database stores data in the form of tables, and a NoSQL database stores data in the form of key-value pairs.
- A DBMS is a software that allows to create, update, and retrieval of data in an organised way. It also provides security to the database.
- Examples of relational DBMS are MySQL, Oracle, Microsoft SQL Server, PostgreSQL, and Snowflake.
- Examples of NoSQL DBMS are MongoDB, Cassandra, DynamoDB, and Redis.



Database Management System

Introduction

Understand the basics of DBMS, explore how it works, and learn about different database architectures used in real-world systems.

1. [DBMS Introduction]
2. [Need for DBMS]
3. [DBMS Architectures (1, 2 and 3 Tier)]

Entity Relationship Model

It is a conceptual model for designing a databases to represent real-world data using entities, attributes, and relationships.

1. ER Model
2. Enhanced ER Model
3. Minimization of ER Diagram
4. Generalization, Specialization and Aggregation
5. Recursive Relationships

Relational Model and Functional Dependencies

Provides a structured way to organize data into tables and how queries are expressed mathematically and logically to retrieve data from relational databases.

1. Relational Model
2. Keys (Candidate, Super, Primary, Alternate and Foreign)
3. Functional Dependency
4. Attribute Closure
5. Equivalence of Functional Dependencies
6. Canonical Cover
7. Anomalies in Relational Model
8. Mapping from ER Model to Relational Model
9. Strategies for Schema design
10. Schema Integration
11. Quiz: ER and Relational Model

Normalization

Important process in database design that helps improve the database's efficiency, consistency, and accuracy.

1. Introduction
2. Normal Forms
3. The Problem of redundancy in Database
4. Dependency Preserving Decomposition
5. Lossless Join Decomposition
6. How to find the Highest Normal Form of a Relation

7. Domain Key normal form
8. Denormalization in Databases
9. Data Replication
10. Quiz: Normal Forms

Relational Algebra and Calculus

Formal query languages used in DBMS to retrieve and manipulate data from relational databases.

1. Introduction
2. Basic Operators
3. Extended Operators
4. Inner Join vs Outer Join
5. Join operation Vs nested query
6. Tuple Relational Calculus
7. Row oriented vs. column oriented data stores

Transactions and Concurrency Control

Ensure reliable execution of database operations and manages simultaneous access without conflicts.

1. Introduction
2. ACID Properties
3. Lock Based Protocol
4. Graph Based Protocol
5. Multiple Granularity Locking
6. Timestamp Ordering Protocols
7. Thomas Write Rule
8. Timestamp and Deadlock Prevention Schemes
9. Log based recovery
10. Types of OLAP Systems
11. Quiz: Transactions and concurrency control

File Organization, Indexing, B and B+ Trees

Method of storing data records in a file so they can be accessed efficiently. It determines how data is arranged, stored, and retrieved from physical storage.

1. File Organization
2. Indexing in Databases
3. SQL queries on clustered and non-clustered Indexes
4. B-Tree
5. B+ Tree
6. Bitmap Indexing
7. Inverted Index

8. Practice questions on B and B+ Trees
9. Quiz: Indexing, B and B+ Trees , File structures

Advanced Topics

Concepts focus on improving performance, scalability, security, and data retrieval across modern systems.

1. RAID
2. Distributed Database System
3. Query Optimization
4. Difference between RDBMS and HBase
5. Challenges of database security
6. Data Management issues in Mobile database
7. Future Works in Geographic Information System
8. Web Information Retrieval | Vector Space Model

Data Warehouse and Data Mining

Store large volumes of historical data and discover useful patterns from it.

1. Data Warehousing
2. Data Warehouse Architecture
3. Data Mining
4. Dimensional Data Modeling
5. Data Marts
6. Apache HBase
7. Apache Hive
8. Architecture and Working of Hive

SQL Tutorial & Interview Questions

SQL is the standard language used to store, retrieve, and manipulate data in relational databases.

1. SQL Tutorial
2. Quiz on SQL
3. SQL Interview Questions
4. PostgreSQL Interview Questions
5. PL/SQL Interview Questions
6. DBMS interview questions

Miscellaneous Topics

Help build a deeper understanding of how databases interact with users, applications, and internal structures.

1. DBMS Interfaces
2. Categories of DBMS Users

3. Data Abstraction and Data Independence
4. Database Objects
5. Impedance Mismatch

DBMS practices questions

- *Database Management Systems | Set 1*
- *Database Management Systems | Set 2*
- *Database Management Systems | Set 3*
- *Database Management Systems | Set 4*
- *Database Management Systems | Set 5*
- *Database Management Systems | Set 6*
- *Database Management Systems | Set 7*
- *Database Management Systems | Set 8*
- *Database Management Systems | Set 9*
- *Database Management Systems | Set 10*
- *Database Management Systems | Set 11*

Quick Links:

1. Last Minutes Notes(LMNs) on DBMS
2. Quizzes on DBMS

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of DBM

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